



P.T.LEE CHENGALVARAYA NAICKER POLYTECHNIC COLLEGE

(A GOVT. AIDED INSTITUTE) / EVK Sampath Salai, Vepery, Chennai – 600 007

DEPARTMENT OF MECHANICAL ENGINEERING

HISTORY OF OUR INSTITUTE

P.T. Lee. Chengalvaraya Naicker, the illustrious founder of this institution, was born in the year 1829. His father, Perumal Naicker, on whom the Government had conferred the title "Sirdar Bahadur", was a subedar-Major in the Madras Army.

His mother Theivayanaiammal, towards whom her son showed a rare respect and devotion, was very pious, orthodox and conservative. Even as a child, Chengalvaraya Naicker was noted for his sweet nature, devotion to God and his parents, and kindness to all around him. As he grew up he developed a keen interest in business which led him to adopt trade and commerce as his profession.

At his father's death, Chengalvaraya Naicker, who was thirty years of age, set up a business of his own and soon found himself a leader of the Indian trading community in Madras. He became a Dubash of Messrs. Shand and Co. and soon amassed a great fortune.

He was very capable and hence successful in business. His heart melted at the sight of the poor and the needy. During the great famine of 1857, his house became a regular relief camp where the necessities of life were distributed to all those who were in need of them, irrespective of their caste and creed. His favourite resort was the great temple in Triplicane. Under his patronage a Sabah was founded with the object of editing and publishing the sacred books of the Vaishnavas.

One of the most notable things in his will is his wish to include initials "P.T.Lee" before his name. He wanted both his father Perumal Naicker and his mother Theivayanaiammal, people to whom he owed his birth and existence, to be remembered in his name as "P.T", besides his pet name "Lee". This was, indeed a highly exemplary and revolutionary act demonstrating his apathy for, and disagreement with, the universally prevalent male chauvinism, reflected in the exclusion of a mother's name in one's initials. We can confidently say that P.T. Lee Chengalvaraya Naicker lived and thought centuries ahead of his times. The great soul passed away in 1874, at the early age of 45.

VISION OF THE INSTITUTE

- To achieve our goal of becoming a model polytechnic institution, we focus on offering both syllabus-oriented classes and job-oriented courses to equip students with the skills needed to excel in engineering fields and adapt to the latest technological developments.

ABOUT MECHANICAL DEPARTMENT

The Department of Mechanical Engineering at P.T. Lee Chengalvaraya Naicker Polytechnic College in Chennai is a cornerstone of the institution's technical education offerings. It provides a diploma program that emphasizes both theoretical knowledge and practical skills, preparing students for careers in industries like manufacturing, automotive, and aerospace.

The department is equipped with modern laboratories, workshops, and tools to ensure hands-on learning. Students are trained in areas such as machine design, thermodynamics, and manufacturing processes. The faculty members are experienced and dedicated to fostering a strong foundation in engineering principles.

The college also focuses on extracurricular activities, including technical clubs and industry visits, to enhance students' exposure to real-world applications. Placement support is provided to help graduates secure jobs in reputed companies.

VISION OF THE DEPARTMENT

To provide quality education to the students of Mechanical Engineering to make them employable and to enhance their technical skills.

MISSION OF THE DEPARTMENT

- To cultivate dynamic learning environment that empowers students with fundamental and advanced mechanical Engineering knowledge.
- To foster creativity and equip them with hands on skills to meet real world challenges while upholding ethical standards and sustainable principles.
- To Support and promote excellence in teaching, research and innovation encouraging collaboration among faculty members and students.
- To develop and maintain the state-of-the-art facilities, Laboratories and technological resources that enhance research and education.

PROGRAM EDUCATIONAL

OBJECTIVES (PEOs)

Graduates of our program will:

- PEO 1: The graduates will be able to apply the learnt concepts in Mechanical Engineering and also in interdisciplinary areas.
- PEO 2: The students will be able to enhance their technical knowledge of inner areas of Mechanical Engineering such as Design, Manufacturing, Maintenance and shop Floor Techniques.
- PEO 3: The students will be trained in skills which required for Employability, Entrepreneurship, Socio-Industrial needs and Value addition in their profession

PROGRAM OUTCOMES (POs)

- **PO-1:** Basic and Discipline specific knowledge: Apply knowledge of basic mathematics, science and engineering fundamentals and engineering specialization to solve the engineering problems.
- **PO-2:** Problem analysis: Identify and analyze well-defined engineering problems using codified standard methods.
- **PO-3 :** Design/ development of solutions: Design solutions for well-defined technical problems and assist with the design of systems components or processes to meet specified needs
- **PO-4:** Engineering Tools, Experimentation and Testing: Apply modern engineering

- tools and appropriate technique to conduct standard tests and measurements.
- **PO-5:** Engineering practices for society, sustainability and environment: Apply appropriate technology in context of society, sustainability, environment and ethical practices.
- **PO-6:** Project Management: Use engineering management principles individually, as a team member or a leader to manage projects and effectively communicate about well-defined engineering activities.
- **PO-7:** Life-long learning: Ability to analyze individual needs and engage in updating in the context of technological changes.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- **PSO 1:** Ability to apply the knowledge of Production and Fabrication for mechanical applications with the help of computer assisted tools.
- **PSO 2:** Ability to implement new ideas and to upgrade the existing products of Mechanical Engineering.

INDUSTRIAL VISIT



The Third Year Mechanical Engineering Students were taken on an Industrial visit to L & T Metro project site Kolappancheri on 29-08-2025 and 30-08-2025



The Third Year B-Sec on Mechanical Engineering Students were taken on an Industrial visit to TCMPT LTD, Ambattur, Chennai-98. on 18-03-2025



The second year Mechanical Students attended the industrial visit to ICF – Integral Coach Factory on Villivakkam, Chennai.



The second year A section students attended the industrial visit to Aavin milk factory on Ambattur, Chennai.



The Second year B section Students attended the industrial visit to Aavin Milk factory on Ambattur, Chennai.

EVENTS CONDUCTED

DIAGONALL CADD, AIEMA



The Branch manager Mr. Boobalan and other technical staffs came to our college to explain about the importance of software, like

CAD-Computer Aided Design Software such as AutoCAD, CATIA, SOLIDWORKS, PTC creo and NX cad.

CAM- Computer Aided Manufacturing software such as MasterCAM, Fusion360.

PETRICORE TECHNOLOGIES, TAMBARAM

The Branch Manager Mr. Pavithan came and explained the importance of Autodesk INVENTOR software, uses and applications.

And also explained the VR- Virtual Reality inclusion in the Realtime world and how to make analysis in virtual.

NAAN MUDHALVAN 3RD SEMESTER



In NAAN MUDHALVAN program conducted by the TANCAM which is located in Thiruvannamiyur.

The course named ADDITIVE MANUFACTURING is conducted by Mr. Venkatesh and Mr. Dinesh.

Softwares such as CATIA and Ultimaker CURA is in that course.

That course makes us to easily understand about the 3D MODELLING and Slicing, The material selection, 3d printer bed speed is mentioned in Ultimaker CURA, Modelling has been done in the CATIA software which is developed by Dassault Systems.

NAAN MUDHALMAN 4TH SEMESTER

From the TANCAM Mr. Anbuselvan arrived to deliver the course IDT- INDUSTRIAL DESIGN TECHNOLOGY in the part of NAAN MUDHALVAN.

Now the 4th semester started till now the introduction class about their Company and the 3D

printer, and some designing modules has been finished.

NSS ACTIVITIES

- AWARENESS ABOUT CYBER SAFETY
- AWARENESS ABOUT CHILD SAFETY
- AWARENESS ABOUT WASTE MANAGEMENT
- AWARENESS ABOUT YOGO AND MEDITATION

GOVERNMENT FUNCTIONS

Attended the KALVIYIL SIRANDHA TAMILNADU in JAWARHAL NEHRU INDOOR STADIUM Which was conducted in the presence of Hon'ble Chief Minister Mr. M.K. Stalin sir.

Attended the STARTUP Awareness program in NANDHAMBAKAM TRADE CENTER which was conducted in the presence of Hon'ble Deputy Chief Minister Mr. Udhayanidhi Stalin sir.

.....STUDENT CORNER.....

NEW INNOVATIONS ON OUR DEPARTMENT.



Mechanical engineering today is evolving rapidly with innovations that are reshaping industries. One major development is the rise of soft robotics, which uses flexible materials to create machines that can safely interact with

humans. This is particularly valuable in healthcare and agriculture, where delicate handling is essential. Another breakthrough is the use of smart materials such as shape-memory alloys and piezoelectric composites, which can sense stress or damage and adapt accordingly, making structures more resilient and efficient.

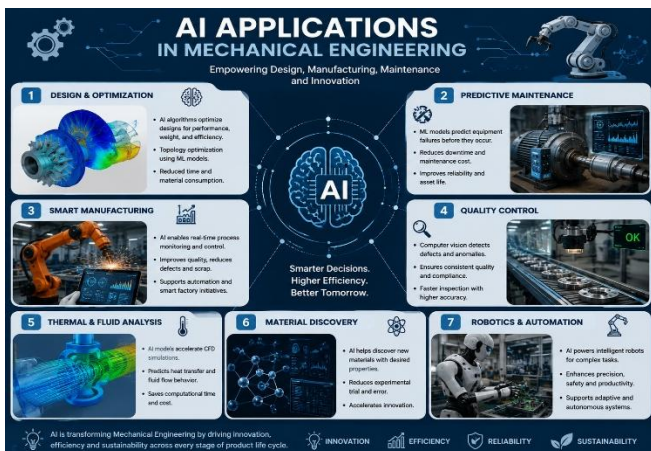
At the same time, **additive manufacturing (3D printing)** has moved beyond prototyping into full-scale production, enabling lightweight designs and reducing material waste significantly. Coupled with this, **digital twins**—virtual replicas of physical assets—are revolutionizing predictive maintenance by allowing engineers to monitor and optimize systems in real time, cutting downtime and costs.

Energy and sustainability are also at the forefront. Mechanical engineers are developing **hybrid cooling systems for EV batteries**, combining liquid cooling with phase-change materials to improve efficiency and safety. There is also a strong push toward **decarbonization technologies**, where mechanical systems are designed with sustainability-first principles, ensuring reduced environmental impact.

Together, these innovations highlight how mechanical engineering is no longer limited to traditional machinery but is deeply integrated with **AI, digital tools, and sustainable practices**. For students and professionals, mastering these areas opens opportunities in industries ranging from aerospace and automotive to energy and offshore engineering.

--K. VASANTH KUMAR 2ND YEAR 'B' SECTION

AI APPLICATIONS IN MECHANICAL DEPARTMENT



Artificial Intelligence (AI) has become a transformative force in mechanical engineering, enabling smarter, faster, and more efficient processes. One of its most significant applications is predictive maintenance, where AI systems analyze sensor data from machines to detect early signs of wear or malfunction. This helps engineers prevent breakdowns, reduce downtime, and extend equipment life. Industries such as automotive, aerospace, and energy rely heavily on these systems to maintain continuous and safe operations.

Another major innovation is **generative design**, which uses AI algorithms to create optimized structures based on specific performance requirements. Engineers input parameters like material type, load, and cost, and AI generates multiple design options that balance strength and efficiency. This approach leads to lightweight, sustainable components and accelerates product development, making it ideal for high-performance sectors like aerospace and automotive manufacturing.

AI also plays a crucial role in **robotics and automation**, where intelligent robots can learn, adapt, and collaborate with humans. These AI-driven systems enhance precision and productivity

in manufacturing plants, performing repetitive or hazardous tasks with minimal supervision. Collaborative robots (cobots) are increasingly used to improve safety and efficiency, reflecting how AI is reshaping industrial workflows.

Finally, **digital twins** and **energy optimization** represent the integration of AI with sustainability. Digital twins create virtual models of physical assets, allowing engineers to simulate performance and predict maintenance needs. Meanwhile, AI-driven energy systems analyze data to minimize power consumption and optimize resource use. Together, these technologies are driving mechanical engineering toward a future that is intelligent, sustainable, and data-driven.

--P. ABISHEK, 2ND YEAR 'A' SECTION

ADVANCED AUTONOMOUS VEHICLES



Advanced autonomous vehicles represent one of the most remarkable achievements in modern mechanical and AI engineering. These vehicles combine sensors, cameras, radar, and LIDAR systems with artificial intelligence to perceive their surroundings and make real-time decisions without human intervention. The mechanical design focuses on precision control systems, adaptive suspension, and energy-efficient powertrains, ensuring smooth operation even in

complex environments.

A key innovation behind autonomous vehicles is sensor fusion, where data from multiple sources is combined to create a comprehensive understanding of the environment. This allows the vehicle to detect pedestrians, traffic signals, and road conditions with high accuracy. Mechanical engineers play a crucial role in designing the structural components that house these sensors, ensuring durability and optimal placement for maximum efficiency. The integration of AI with mechanical systems has led to vehicles capable of adaptive cruise control, lane keeping, and collision avoidance—all essential for safe autonomous driving.

Another important aspect is **vehicle-to-everything (V2X) communication**, which enables autonomous cars to interact with other vehicles, infrastructure, and cloud systems. This connectivity allows real-time updates on traffic, weather, and road hazards, improving decision-making and reducing accidents. Mechanical engineers collaborate with AI specialists to design robust communication modules and ensure that mechanical systems respond instantly to digital commands. The synergy between mechanical precision and AI intelligence is what makes these vehicles truly autonomous.

Waymo currently leads the global autonomous driving industry with an impressive ADAS (Advanced Driver Assistance System) score of 87.9 out of 100, according to Telemetry's 2026 Automated Driving Systems Scorecard. This score reflects Waymo's dominance in technology readiness, safety, and commercial scalability across multiple U.S. cities.

--B.S. MURALI KRISHNA 2ND YEAR 'B' SECTION

INDUSTRY 5.0



Industry 5.0, however, emphasizes the *human-centric* side of technology. Instead of just automation, it's about collaboration between humans and intelligent machines. The goal is not only productivity but also sustainability, resilience, and personalization. For example, cobots (collaborative robots) are designed to work safely alongside humans, while AI systems help engineers design greener, more adaptive solutions.

In mechanical engineering, Industry 5.0 means integrating **AI, advanced materials, and sustainable energy systems** with traditional mechanical design. It's about creating smarter products that are environmentally responsible and tailored to user needs. Think of autonomous vehicles optimized for safety and sustainability, or manufacturing systems that adapt in real time to reduce carbon emissions.

So, to put it simply: **Industry 4.0 = automation and efficiency; Industry 5.0 = human-machine collaboration and sustainability.** For your career path, mastering tools like **digital twins, AI-driven design, and sustainable mechanical systems** will align you perfectly with Industry 5.0 expectations.

--H.HARRY MECHANICAL 2ND YEAR 'A' SECTION

திடக் கழிவு மேலாண்மை (Solid Waste Management)



திடக் கழிவு மேலாண்மை (Solid Waste Management) என்பது சுற்றுச்சூழல் பாதுகாப்பில் மிக முக்கியமான பங்காற்றும் துறை. மனிதர்கள், தொழிற்சாலைகள் மற்றும் மருத்துவமனைகள் போன்ற இடங்களில் உருவாகும் பாழ்பொருட்களை சரியான முறையில் சேகரித்து, பிரித்து, மறுசுழற்சி செய்து, மீண்டும் பயன்பாட்டிற்கு கொண்டு வருவது இதன் நோக்கம். இதன் மூலம் மாசு குறைந்து, இயற்கை வளங்கள் பாதுகாக்கப்படுகின்றன.

முதலில், கழிவுகளை மூலத்தில் பிரித்தல் (Segregation at Source) மிக அவசியம். உயிரியல் கழிவுகள் (Biodegradable) மற்றும் உயிரியல் அல்லாத கழிவுகள் (Non-biodegradable) என பிரித்து சேகரிக்க வேண்டும். உயிரியல் கழிவுகள் உரமாக மாற்றப்படலாம்; உயிரியல் அல்லாதவை மறுசுழற்சி தொழில்நுட்பம் மூலம் புதிய பொருட்களாக உருவாக்கப்படலாம்.

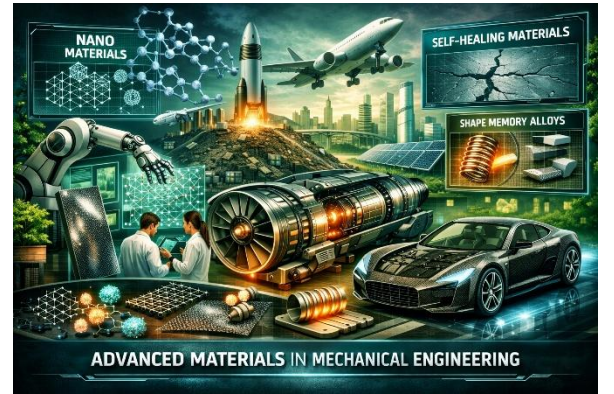
அடுத்ததாக, **மறுசுழற்சி (Recycling)** மற்றும் **மீள்பயன்பாடு (Reuse)** ஆகியவை திடக் கழிவு மேலாண்மையின் முக்கியமான கட்டங்கள். பிளாஸ்டிக், காகிதம், உலோகம் போன்ற பொருட்களை மறுசுழற்சி செய்வதன்

மூலம் புதிய பொருட்கள் தயாரிக்கலாம். இதனால் இயற்கை வளங்கள் குறைவாக பயன்படுத்தப்படுகின்றன மற்றும் சுற்றுச்சூழல் மாசு குறைகிறது.

இறுதியாக, **சுற்றுச்சூழல் விழிப்புணர்வு (Environmental Awareness)** மிக அவசியம். மக்கள் தங்கள் தினசரி வாழ்க்கையில் கழிவுகளை குறைக்கும் பழக்கத்தை வளர்த்துக்கொள்ள வேண்டும். நகராட்சிகள் மற்றும் தொழிற்சாலைகள் திடக் கழிவு மேலாண்மைக்கு நவீன தொழில்நுட்பங்களை (Smart Waste Systems, IoT-based Monitoring) பயன்படுத்தி பசுமையான நகரங்களை உருவாக்க முடியும். இதுவே நிலைத்த வளர்ச்சிக்கான அடிப்படை வழி.

--DEVESH KUMAR MECHANICAL 3RD YEAR 'A' SECTION

மேம்பட்ட பொருட்கள் (Advanced Materials)



மேம்பட்ட பொருட்கள் (Advanced Materials) என்பது நவீன தொழில்நுட்ப வளர்ச்சியின் அடிப்படை தூணாகும். இவை இயந்திர பொறியியல், வான்வழி, மருத்துவம், மின்னணு மற்றும் ஆற்றல் துறைகளில் பெரும் மாற்றத்தை ஏற்படுத்துகின்றன. மேம்பட்ட பொருட்கள் என்பது அதிக வலிமை, குறைந்த எடை, அதிக வெப்பத்திற்கும் அழுத்தத்திற்கும் எதிர்ப்பு, மற்றும் நீண்ட ஆயுட்காலம் கொண்ட பொருட்கள் ஆகும். இவை பாரம்பரிய

உலோகங்கள் மற்றும் பாலிமர் பொருட்களை விட சிறந்த செயல்திறனை வழங்குகின்றன.

முதன்மையாக, நானோ பொருட்கள் (Nanomaterials) மற்றும் கலவை பொருட்கள் (Composite Materials) ஆகியவை முக்கிய பங்காற்றுகின்றன. நானோ அளவிலான துகள்கள் பொருட்களின் இயற்பியல் மற்றும் வேதியியல் பண்புகளை மாற்றி, அதிக வலிமை மற்றும் நீடித்த தன்மையை வழங்குகின்றன. கலவை பொருட்கள், இரண்டு அல்லது அதற்கு மேற்பட்ட பொருட்களை இணைத்து உருவாக்கப்படுவதால், விமானங்கள், வாகனங்கள் மற்றும் ரோபோடிக்ஸ் துறைகளில் அதிக பயன்பாட்டை பெற்றுள்ளன.

அடுத்ததாக, சுய-மருத்துவ பொருட்கள் (Self-Healing Materials) மற்றும் வடிவ நினைவுப் பொருட்கள் (Shape Memory Alloys) போன்றவை நவீன பொறியியல் பயன்பாடுகளில் முக்கியமானவை. இவை தானாகவே சேதத்தை சரிசெய்யும் திறன் கொண்டவை அல்லது வெப்பம் மற்றும் அழுத்தம் மாற்றத்தால் தங்கள் வடிவத்தை மீண்டும் பெறும் தன்மை கொண்டவை. இதனால் இயந்திரங்கள் மற்றும் கட்டமைப்புகள் நீண்ட காலம் செயல்பட முடிகிறது.

இறுதியாக, சுற்றுச்சூழல் நட்பு பொருட்கள் (Eco-Friendly Materials) மற்றும் செயற்கை நுண்ணறிவு (AI)-ஆல் வடிவமைக்கப்பட்ட பொருட்கள் எதிர்கால வளர்ச்சியின் திசையை நிர்ணயிக்கின்றன. இவை ஆற்றல் சேமிப்பு, மறுசுழற்சி மற்றும் நிலைத்த வளர்ச்சிக்கு வழிவகுக்கின்றன. மேம்பட்ட பொருட்கள் மனித வாழ்க்கையை பாதுகாப்பாகவும், திறமையாகவும் மாற்றும் புதிய தொழில்நுட்ப யுகத்தின் அடையாளமாகும்.

இவற்றில் **நானோ பொருட்கள் (Nanomaterials)** முக்கிய பங்கு வகிக்கின்றன. நானோ அளவிலான துகள்கள் பொருட்களின் இயற்பியல் மற்றும் வேதியியல் பண்புகளை மாற்றி, அதிக வலிமை மற்றும் நீடித்த தன்மையை வழங்குகின்றன. இதனால் விமானங்கள், வாகனங்கள் மற்றும் ரோபோடிக்ஸ் துறைகளில் அதிக செயல்திறன் பெறப்படுகிறது.

--S. BALASUBRAMANI MECHANICAL 3RD YEAR 'A' SECTION

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MECHANICAL DEPARTMENT 2ND YEAR

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<https://mechcnpt.github.io/cnptmech/>

COLLEGE ADDRESS:

P.T.Lee Chengalvaraya Naicker Polytechnic College

[Formerly C.N. Technical Institute and C.N.
Polytechnic]

(A Govt. Aided Institution – Estd.: 1939)

2&3, E.V.K. Sampath Salai, Vepery, Chennai – 600 007.

Tamil Nadu, India.

WORKING HOURS

Full Time : Monday - Friday: 09:15 AM - 05:00 PM

Part Time : Monday - Saturday: 06:00 PM - 09:00 PM

Weekends : Closed

